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Sleep patterns before and after competition: A real-world examination of elite athletes

Jian En Sim[a,b], Joshua Leota[a,b], Luis Mascaro[a,b], Daniel Hoffman[a,b] and Elise R. Facer-Childs[a,b,c]

[a]School of Psychological Sciences, Monash University, Melbourne, Australia; [b]Turner Institute for Brain and Mental Health, Monash University, Melbourne, Australia; [c]High Performance Department, St Kilda Football Club, Australian Football League, Melbourne, Australia

ABSTRACT

Good sleep before and after competitions is crucial to cognitive, physiological performance and recovery. Yet, elite athletes face a unique set of challenges when acquiring good sleep before and after competitions, and indeed commonly report sleep problems when it matters most. This study examined the sleep of elite athletes before and after competition compared to before and after free days. A total of 1808 unique nights of actigraphy ($n = 1495$) and sleep diary ($n = 1335$) data from elite Australian Football League and National Rugby League male athletes ($N = 85$, M-age = 24.4 ± 3.6) were collected and analysed using multi-level mixed models. On nights before competitions, athletes advanced sleep timings ($p < .001$, d = 0.63) and increased total sleep time ($p < .001$, d = 0.65) compared to nights before free days. On nights after competitions, athletes delayed sleep timings ($p < .001$, d = 1.64), reduced total sleep time ($p < .001$, d = 1.28), and had significantly worse quality sleep ($p < .001$, d = 0.71) compared to nights after free days. Sleep was especially worse following night competitions. While elite athletes and organisations may be implementing effective sleep strategies to optimise sleep the night before competitions, strategies to improve sleep after competitions may be lacking. We discuss potential factors contributing to this asymmetry and propose areas for research moving forward.

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Match day; Australian Football League (AFL); National Rugby League (NRL); performance; recovery; health

1. Introduction

Sleep plays an integral role in an elite athlete's health, well-being, and performance (Walsh et al., 2021). The importance of sleep is well recognised in professional sports, with a significant number of teams and athletes employing sleep coaches to improve performance and recovery (Gómez-Aldana, 2023; N. Jones, 2023). Despite efforts to leverage the science of sleep, sleep problems in elite athletes remain prevalent, particularly during the critical nights prior to and following competitions (Erlacher et al., 2011; Juliff et al., 2015; O'Donnell et al., 2018). Here, we examined the sleep of elite National Rugby League (NRL) and Australian Football League (AFL) male athletes to shed light on how various pressures of competitive matches may be influencing sleep.

1.1. Sleep before competitions

Good sleep (defined hereafter as sufficient sleep quantity and quality) has been associated with greater overall athletic performance on various metrics, including higher accuracy, faster reaction times, and greater physical endurance (Mah et al., 2011; Roberts et al., 2019; Schwartz & Simon, 2015). Good sleep is also linked to improved cognitive performance and psychological resilience among elite athletes (Schwartz & Simon, 2015). However, obstacles exist that may prevent athletes from achieving good sleep. Elite athletes cite difficulty initiating sleep as a primary factor contributing to poor sleep the night before competitions (Ghorbanzadeh & Bayar, 2013). This can be due to competitions bearing significant psychological stress, anxiety, and excitement for athletes, which may in turn heighten pre-sleep hyperarousal, a key process involved in insomnia (Gupta et al., 2017). Poor sleep can affect an athlete's physiology, resulting in reduced immune function, increased injury risk, and greater fatigue (Fullagar et al., 2015; Milewski et al., 2014; Sargent et al., 2014). Poor sleep can also affect an athlete's cognitive performance, mood, and motivation to train (Lastella et al., 2020; Robson-Ansley et al., 2009; Taylor et al., 1997).

Several studies have reported disruptions to subjective sleep the night before elite competitions. For example, 78% of elite female gymnasts reported poor quality sleep the night before an international competition (Silva & Paiva, 2016). 66% of German athletes from various sports reported experiencing poor sleep the night before a sports event (Erlacher et al., 2011). Similarly, 64% of Australian athletes reported worse sleep the night before an important competition (Juliff et al., 2015). Interestingly, most athlete studies that collect objective sleep data via actigraphy reported no differences to sleep quality and duration on nights before competitions (Gupta et al., 2017). In fact, two studies of elite Australian Rules

CONTACT Elise R. Facer-Childs elise.facer-childs@monash.edu Turner Institute for Brain and Mental Health, School of Psychological Sciences, 18 Innovation Walk, Monash University, Clayton VIC 3800, Australia

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Footballers reported increased sleep duration prior to competition (L. Richmond et al., 2004; L. K. Richmond et al., 2007).

1.2. Sleep after competitions

Good sleep following vigorous exercise inherent to competitive sport has restorative benefits and is critical to promoting faster recovery (Bonnar et al., 2018; Halson, 2014). The vast majority of human growth hormone production occurs during sleep (Redwine et al., 2000). Human growth hormones promote the growth and repair of muscle tissue, thereby accelerating recovery (Venter, 2012). However, the physical and cognitive demands of elite competitions may disrupt subsequent sleep episodes. Elevated adrenaline and cortisol levels after competitions is associated with heightened pre-sleep arousal (Garde et al., 2012). and muscle pain and soreness is associated with increased sleep disturbances (Taylor et al., 1997). Rumination about one's performance, particularly following losses, may lead to later sleep onset timing (Fullagar et al., 2016).

Caffeine and alcohol consumption may also interfere with sleep. Caffeine, known for its performance-enhancing capabilities, is often consumed by elite athletes before competitions to boost alertness and performance (Pickering & Kiely, 2019; Santos et al., 2014). However, caffeine consumption, particularly later in the day, can decrease sleep quality and quantity (Clark & Landolt, 2017; Dunican et al., 2018; McLellan et al., 2016). Alcohol is sometimes consumed after competitions as a reward or as part of sporting culture (e.g., the after-match function) (Barnes, 2014) and is associated with reduced rapid eye-movement (REM) sleep duration and increased sleep disturbances (Park et al., 2015; Roehrs & Roth, 2001). Other post-competition commitments such as team meetings, media availability, injury management and travel may also contribute to poor sleep (Sargent et al., 2022).

Later competition timings (i.e., evening versus day) may also delay sleep onset time. Competitions later in the day result in higher night-time arousal and core body temperature, which may contribute to difficulties initiating sleep (Sargent & Roach, 2016), while bright light exposure – especially during competitions scheduled at night – may delay an athlete's circadian phase (O'Donnell et al., 2018). Poor sleep can also increase pro-inflammatory cytokines, which impedes muscle recovery and repair from damage (Vitale et al., 2019). This may subsequently lead to overtraining syndrome, which can occur when an athlete does not sufficiently recovery from strenuous exercise (Jürimäe et al., 2004).

Table 1. Means and standard deviations (SD) of general characteristics of participants.

Variable	Sport Type	Mean	SD
Age	AFL	24.29	3.54
	NRL	24.43	3.63
Weight (kg)	AFL	87.16	8.06
	NRL	98.25	11.34
Height (cm)	AFL	189.39	7.33
	NRL	186.61	5.97
Midsleep (hh:mm)	AFL	03:24	32 min
	NRL	03:31	39 min

Note: AFL = Australian Football League, NRL = National Rugby League.

Indeed, empirical evidence supports the theoretical view that sleep is severely disrupted following elite competitions (Gupta et al., 2017). For example, contact sport studies (rugby league, rugby union, AFL, soccer) reported significant decreases to sleep quality (Fowler et al., 2014; Fowler et al., 2016; Fullagar et al., 2016; Richmond et al., 2004) and sleep duration (Caia et al., 2017; Fowler et al., 2014; Fowler et al., 2016; Fullagar et al., 2016; Fullagar et al., 2016; Richmond et al., 2004; Shearer et al., 2015), and increases to sleep onset (Sargent et al., 2022) and sleep onset latency (Fullagar et al., 2016) on nights following competitions.

1.3. The current research

For elite athletes to perform at the highest level, every aspect of preparation and recovery is important, including an athlete's ability to sleep well. Given the many obstacles associated with the psychological and physical stressors of elite competitions, it is plausible that elite athletes' sleep will be degraded the nights before and after competitions, and especially degraded on nights following evening competitions. On the other hand, the recent increase in sleep recommendations from sleep researchers, experts, and consultants have led to elite athletes becoming more aware of the crucial role sleep plays on and off the field (Kroshus et al., 2019; Singh et al., 2021). Athletes may therefore now attempt to extend their sleep opportunity the nights before and after competitions to optimise performance and to facilitate recovery (Li et al., 2018; Richmond et al., 2004; Shearer et al., 2015).

The aim of the current study is to better understand how elite contact sports athletes sleep before and after competitions in a free-living and ecologically valid setting. Specifically, we tracked both objective and subjective sleep measures in a group of elite Australian male contact sports athletes, providing a real-world examination of how sleep is impacted during their competition season.

2. Methods

2.1. Participants

Elite contact sports athletes were recruited from an elite AFL team and an elite NRL team. Data from 85 male athletes (AFL = 51, NRL = 34) were collected. Descriptive statistics of the participants from each sport type are reported in Table 1. Participation in the study was voluntary and written informed consent was obtained prior to any data collection. The study was approved by Monash University's Human Research Ethics Committee, approval number #26361.

2.2. Procedure

Data from 85 athletes were collected across 3 timepoints (AFL = 2 timepoints, NRL = 1 timepoint) for a total of 1,808 (actigraphy = 1495, sleep diary = 1335) nights of sleep data. Each timepoint ran for 2 to 3 weeks. Within each timepoint, athletes engaged in at-home sleep monitoring via wrist-worn actigraphy and electronic sleep diaries. Days of the week were categorised into 3 event types: free days (n = 633), competition days

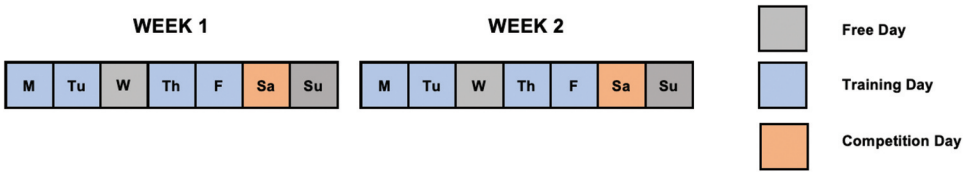

Figure 1. Typical schedule for the athletes during a single timepoint of data collection. Schedules may change slightly from week to week depending on location and date of competitions.

($n = 186$), and training days ($n = 889$). Figure 1 illustrates a typical schedule for the athletes in a single timepoint of data collection.

2.3. Measures and materials

Sleep variables were measured objectively and subjectively, using GENEActiv actigraphy devices (ActivInsights, Huntingdon, United Kingdom) and daily electronic sleep diaries, respectively. Actigraphic measures of sleep have been validated against polysomnography, showing high sensitivity and accuracy (Aili et al., 2017; Sadeh, 2011). The GENEActiv actigraphy devices used in the current study have been validated against sleep detection measures, demonstrating consistency with other actigraphy devices (Pavey et al., 2016; Salagaras et al., 2021; Stone et al., 2020). Data were collected in 60-second epochs at 30 Hz frequency, with sleep detection settings set to medium. Sleep variables measured include sleep onset, sleep offset, wake after sleep onset (WASO), total sleep time (TST), and sleep efficiency (SE). Sleep diary questions were derived from the consensus sleep diary (Carney et al., 2012). Additional measures on state pre-sleep arousal, caffeine intake, and alcohol intake were adjoined to the sleep diaries. State pre-sleep arousal was measured via a brief state version of the Pre-Sleep Arousal Scale (PSAS) (Nicassio et al., 1985). The PSAS also measured excitement and nervousness before a game. Athletes answered on a five-point Likert scale (1 = *Not at all* to 5 = *Extremely*). Higher scores represent a higher level of state pre-sleep arousal. The participants were also asked to subjectively rate their previous night's sleep quality on a scale from 0 (poor sleep quality) to 10 (excellent sleep quality).

2.4. Data analysis

Data from the actigraphy devices were exported using the GENEActiv software (Version 3.3). Cicada toolbox (v0.10.4 (beta)) open-source software was used for analysing actigraphy data in MATLAB R2019b (MathWorks, Natick, MA) (Wassing, 2023). Responses from sleep diaries were used to identify the sleep episodes for actigraphy analysis. Output from Cicada and responses from sleep diaries were then converted into a data set and analysed using the R programming language with the lme4 and emmeans packages (R Core Team, 2021).

2.4.1. Sleep before competition

To examine how athletes slept the night *before* a competition, a linear mixed model with 1 fixed factor (next day type) and 2 random factors (participant identifier and current day type) was used (see Table 2 for model information). The *current day* variable was included as a random factor to control for variance related to what happened that day (e.g., controls for the influence of a training day if a training day sometimes precedes a competition day). The *next day* fixed variable was dummy coded, treating competition days as a reference and thus comparing competition days to free days and training days. Because AFL and NRL are sports with different schedules and demands on the body, the sport type was included as a fixed factor covariate to account for sport-related differences.

2.4.2. Sleep after competition

To examine how athletes sleep the night *after* a competition, a linear mixed model with 1 fixed factor (current day type) and 2 random factors (participant identifier and next day type) was used (see Table 2). The *next day* variable was included as a random factor to control for variance related to what the athlete expects may happen the following day. The *current day* fixed variable was dummy coded, treating competition days as a reference.

To examine sleep differences between the nights following day and evening competitions, a linear mixed model with 1 fixed factor (game time) and 2 random factors (participant identifier and next day) was used (see Table 2). In our data, day competition start-times ranged from 16:05 hr to 16:35 hr and evening competition start-times ranged from 19:25 hr to 19:50 hr.

3. Results

3.1. Sleep before competition

Table 3 summarises the effect of next day type on all sleep variables. On nights before a competition day, athletes went to bed earlier (22:56 hr ± 49.2 min), woke up later (8:12 hr ± 58.6 min), and slept for longer (7:57 hr ± 60.9 min), compared to nights before a free day (Onset: $M = 23{:}55$ hr ± 107.4 min; Offset: 8:00 hr ± 81.4 min; TST: 6:55 hr ± 95.4 min) and nights before a training day (Onset: 22:59 hr ± 62.9 min; Offset: 7:22 hr ± 57.1 min; TST: 7:14 hr ± 60.5 min). Self-reported excitement and nerves about the upcoming game were significantly higher on nights before a competition day (Excitement: 1.90 ± 1.22; Nerves: 1.39 ± 0.68) compared to nights before a free day (Excitement: 1.56 ± 0.99; Nerves: 1.30 ± 0.61) and nights before a training day (Excitement: 1.61 ± 1.03; Nerves: 1.27 ± 0.55). Subjective sleep quality was higher on nights before competition days (6.94 ± 1.71) compared to nights before free days (6.28 ± 1.93) and training days (6.56 ± 1.56). Finally, alcohol consumption was significantly lower on nights before a competition (1.01 ± 0.08) compared to nights before a free day

Table 2. Summary of models analysed for sleep before competitions, sleep after competitions, and sleep differences between nights following day and evening competitions.

Analysis	Dependent Variables	Fixed Effect(s)	Random Effect(s)
To examine how athletes slept the night before a competition	- Sleep variables (onset, offset, TST, SE, WASO) - State cognitive and somatic arousal - Nerves and excitement before games - Caffeine and alcohol intake - Subjective sleep quality	- Next day type (competition vs. training vs. free) - Sport type (AFL vs NRL)	- Participant ID - Current day type (competition vs. training vs. free)
To examine how athletes slept the night after a competition	- Sleep variables (onset, offset, TST, SE, WASO) - State cognitive and somatic arousal - Caffeine and alcohol intake - Subjective sleep quality	- Current day type (competition vs. training vs. free) - Sport type (AFL vs NRL)	- Participant ID - Next day type (competition vs. training vs. free)
To examine sleep differences between nights following day and evening competitions	- Sleep variables (onset, offset, TST, SE, WASO) - State cognitive and somatic arousal - Caffeine and alcohol intake - Subjective sleep quality	- Game time (day vs evening) - Sport type (AFL vs NRL)	- Participant ID - Next day type (competition vs. training vs. free)

(1.24 ± 0.85). Figure 2 shows the comparison of means for nights before competition days, nights before free days, and nights before training days.

3.2. Sleep after competition

Table 4 summarises the effect of day type on all sleep variables. For nights after competition, sleep was negatively impacted across nearly all variables (Onset: 25:09 hr ± 129.9 min; Offset: 8:05 hr ± 88.2 min; TST: 5:51 hr ± 107.1 min; SE: 80.28 ± 10.90; WASO: 14.97 ± 9.63) compared to nights after a free day (Onset: 23:04 hr ± 70.2 min: Offset: 7:27 hr ± 63.2 min; TST: 7:14 hr ± 65.5 min; SE: 83.24 ± 7.01; WASO: 13.03 ± 6.17) and nights after a training day (Onset: 23:01 hr ± 59.9 min: Offset: 7:40 hr ±66.3 min; TST: 7:27 hr ± 63.9 min; SE: 82.99 ± 6.99; WASO: 13.28 ± 6.09). Cognitive and somatic pre-sleep arousal also increased the night after competition (Cognitive: 1.40 ± 0.82; Somatic: 1.40 ± 0.76) compared to nights after free days (Cognitive: 1.34 ± 0.67; Somatic: 1.36 ± 0.67) and nights after training days (Cognitive: 1.23 ± 0.55; Somatic: 1.24 ± 0.52). Subjective sleep quality was lower on nights after competitions (5.41 ± 2.27) compared to nights after free days (6.54 ± 1.56) and training days (6.56 ± 1.59). Finally, caffeine consumption was significantly higher (2.15 ± 0.82) on competition days compared to free days (1.95 ± 0.71) and training days (2.02 ± 0.70). Figure 3 shows the comparison of means for nights after competition days, nights after free days, and nights after training days.

Table 5 summarises the effect of game time on all sleep variables. For nights following an evening competition, athletes went to bed later (Sleep Diary Onset: 25.21 ± 1.72) and slept for shorter (Actigraphy TST: 5.69 ± 1.75) compared to nights following a day competition (Sleep Diary Onset: 24.46 ± 1.76; Actigraphy TST: 6.17 ± 1.82). Figure 4 shows the comparison of means for nights after evening competitions and day competitions.

4. Discussion

Sleep plays a crucial role in an elite athlete's health, well-being, and performance, and having sufficient good quality sleep is especially important during the competitive season. However, the existing literature remains unclear on how elite athletes are sleeping before and after competitions; a question of particular relevance today, given the increased awareness of the importance of sleep for performance and recovery. The current study examines how elite contact sports athletes sleep before and after competitions. Our findings were mixed but surprisingly specific. Elite AFL and NRL male athletes experience significantly poorer sleep on nights after, but not before, competitions.

4.1. Sleep before competitions

Sleep was not negatively affected the night before competition when compared to the night before free or training days. Athletes slept earlier and for longer, which is consistent with past research (Richmond et al., 2004; Shearer et al., 2015). These results suggest athletes may be aware of the importance of having sufficient rest before a competition and may attempt to increase their sleep opportunity to boost performance. Support staff may likewise recognise the benefits of good sleeping habits, and therefore better educate athletes on the importance of sufficient rest. Athletes may also be aware of the difficulties associated with sleeping before competition, such as difficulties initiating sleep due to increased psychological and physiological arousal (Gupta et al., 2017), and therefore compensate by going to bed earlier. Indeed, we found that while athletes slept longer, they also experienced more excitement and nerves the night before competition. Mindfulness meditation and instrumental emotion regulation strategies may help athletes reduce the arousal experienced the night before competitions (Lane et al., 2011; Mehrsafar et al., 2019). Although sleep quantity was increased, objective sleep quality (e.g., sleep efficiency and WASO) was not significantly different between nights before competitions days and free or training days. However, self-reported subjective sleep quality was significantly higher on nights before competition days compared to free or training days. Athletes may have perceived the longer sleep times before competitions as better sleep quality, even though objective measures did not report any differences.

Table 3. Linear mixed model analysis of sleep variables comparing between nights before competition days, nights before free days, and nights before training days.

Variable	Fixed Effects	Estimate	St. Error	p	d
Sleep Onset (SD)	Intercept	23.48	0.65	<.001***	
	Before Free	0.34	0.10	.001**	0.63
	Before Training	−0.18	0.11	.009*	0.11
	Sport Type	−0.02	0.13	.86	
Sleep Onset (Acti)	Intercept	23.68	0.58	<.001***	
	Before Free	0.46	0.10	<.001***	0.63
	Before Training	−0.20	0.10	.06	
	Sport Type	−0.05	0.15	.72	
Sleep Offset (SD)	Intercept	7.96	0.16	<.001***	
	Before Free	−0.29	0.09	.001**	0.11
	Before Training	−1.00	0.09	<.001***	0.70
	Sport Type	0.44	0.13	<.001***	
Sleep Offset (Acti)	Intercept	8.12	0.16	<.001***	
	Before Free	−0.25	0.09	.004**	0.16
	Before Training	−1.10	0.09	<.001***	0.87
	Sport Type	0.47	0.13	<.001***	
TST (SD)	Intercept	8.04	0.62	.004**	
	Before Free	−0.61	0.12	<.001***	0.65
	Before Training	−0.84	0.12	<.001***	0.70
	Sport Type	0.46	0.13	<.001***	
TST (Acti)	Intercept	7.28	0.48	.002**	
	Before Free	−0.63	0.10	<.001***	0.72
	Before Training	−0.70	0.10	<.001***	0.70
	Sport Type	0.32	0.12	.01*	
SE (SD)	Intercept	83.91	2.48	<.001***	
	Before Free	−1.19	0.79	.30	
	Before Training	−0.48	0.83	.09	
	Sport Type	0.56	1.23	.66	
SE (Acti)	Intercept	82.50	1.21	<.001***	
	Before Free	−0.33	0.58	.57	
	Before Training	1.37	0.60	.02*	0.11
	Sport Type	−1.47	1.09	.18	
WASO (SD)	Intercept	1.54	0.30	<.001***	
	Before Free	0.33	0.23	.15	
	Before Training	0.09	0.23	.71	
	Sport Type	−0.36	0.31	.25	
WASO (Acti)	Intercept	13.15	0.98	<.001***	
	Before Free	0.10	0.51	.85	
	Before Training	−0.85	0.52	.10	
	Sport Type	1.51	0.93	.11	
PSAS (Cognitive)	Intercept	1.26	0.08	<.001***	
	Before Free	0.08	0.06	.21	
	Before Training	0.03	0.06	.61	
	Sport Type	−0.02	0.08	.77	
PSAS (Somatic)	Intercept	1.27	0.08	<.001***	
	Before Free	0.08	0.06	.19	
	Before Training	0.06	0.06	.31	
	Sport Type	−0.08	0.07	.26	
PSAS (Excite.)	Intercept	1.93	0.16	<.001***	
	Before Free	−0.20	0.07	.002**	0.20
	Before Training	−0.20	0.06	.001**	0.27
	Sport Type	−0.22	0.21	.29	
PSAS (Nerves)	Intercept	1.35	0.08	<.001***	
	Before Free	−0.12	0.05	.01*	0.17
	Before Training	−0.14	0.05	.002**	0.21
	Sport Type	−0.04	0.10	.71	
Subj. Sleep Quality	Intercept	6.29	0.43	<.001***	
	Before Free	−0.36	0.14	.01*	0.35
	Before Training	−0.36	0.15	.01*	0.24
	Sport Type	0.39	0.24	.11	
Caffeine	Intercept	1.92	0.12	<.001***	
	Before Free	0.02	0.05	.69	
	Before Training	−0.02	0.05	.071	
	Sport Type	0.16	0.12	.19	
Alcohol	Intercept	1.06	0.06	<.001***	
	Before Free	0.19	0.05	<.001***	0.32
	Before Training	−0.07	0.05	.13	
	Sport Type	0.02	0.03	.50	

Note: *$p < .05$, **$p < .01$, ***$p < .001$. SD = Sleep diary data, Acti = Actigraphy data.

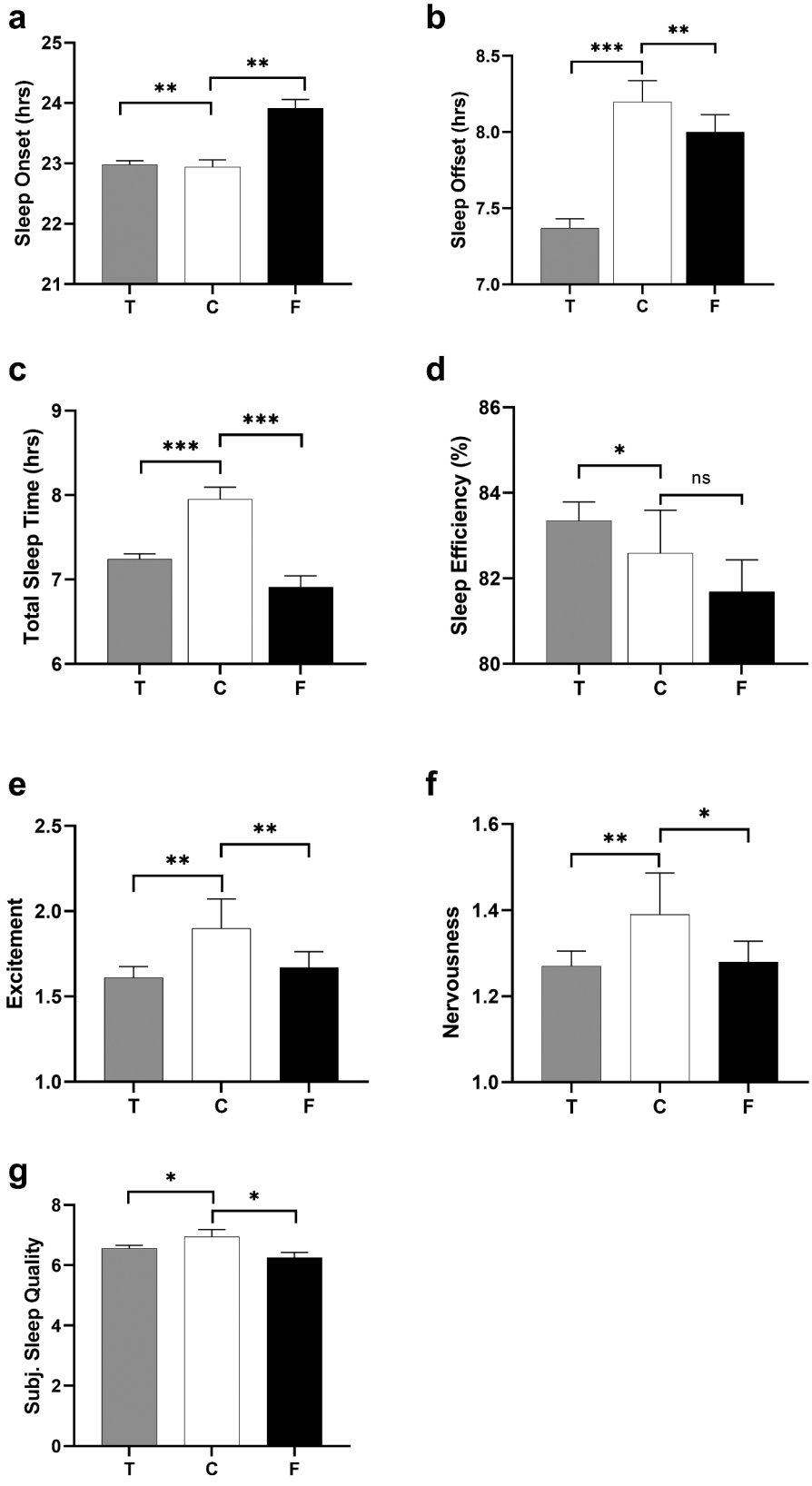

Figure 2. Bar graphs comparing the means for sleep *before* competition days, before free days, and before training days for (a) sleep onset; (b) sleep offset; (c) total sleep time; (d) sleep efficiency; (e) pre-sleep excitement for the upcoming match; (f) pre-sleep nervousness for the upcoming match; (g) subjective sleep quality. Error bars represent 95% CI. *$p < .05$, **$p < .01$, ***$p < .001$, T = before training day, C = before competition day, F = before free day.

Table 4. Linear mixed model analysis of sleep variables comparing nights after competition days, nights after free days, and nights after training days.

Variable	Fixed Effects	Estimate	St. Error	p	d
Sleep Onset (SD)	Intercept	24.81	0.20	<.001***	
	After Free	−1.76	0.11	<.001***	1.64
	After Training	−2.05	0.10	<.001***	2.03
	Sport Type	−0.03	0.13	.82	
Sleep Onset (Acti)	Intercept	24.90	0.25	<.001***	
	After Free	−1.56	0.12	<.001***	1.48
	After Training	−1.83	0.11	<.001***	1.74
	Sport Type	−0.06	0.15	.70	
Sleep Offset (SD)	Intercept	7.63	0.33	<.001***	
	After Free	0.01	0.10	.89	
	After Training	−0.31	0.09	<.001***	0.27
	Sport Type	0.44	0.13	.001**	
Sleep Offset (Acti)	Intercept	7.80	0.36	<.001***	
	After Free	−0.05	0.10	.62	
	After Training	−0.34	0.09	<.001***	0.36
	Sport Type	0.47	0.13	<.001***	
TST (SD)	Intercept	6.33	0.28	<.001***	
	After Free	−1.86	0.12	<.001***	1.28
	After Training	1.80	0.11	<.001***	1.57
	Sport Type	0.46	0.13	<.001***	
TST (Acti)	Intercept	5.90	0.25	<.001***	
	After Free	−1.38	0.11	<.001***	1.11
	After Training	1.42	0.11	<.001***	1.35
	Sport Type	0.32	0.12	.01*	
SE (SD)	Intercept	78.47	1.17	<.001***	
	After Free	7.08	0.79	<.001***	0.71
	After Training	6.91	0.75	<.001***	0.78
	Sport Type	0.61	1.23	.63	
SE (Acti)	Intercept	81.18	1.11	<.001***	
	After Free	2.16	0.67	<.001***	0.37
	After Training	2.67	0.62	<.001***	0.35
	Sport Type	−1.42	1.09	.19	
WASO (SD)	Intercept	2.06	0.30	<.001***	
	After Free	−0.50	0.22	.02*	0.24
	After Training	−0.30	0.22	.17	
	Sport Type	−0.40	0.30	.20	
WASO (Acti)	Intercept	14.17	0.90	<.001***	
	After Free	−1.71	0.58	.003**	0.28
	After Training	−1.91	0.54	<.001***	0.25
	Sport Type	1.44	0.93	.13	
PSAS (Cognitive)	Intercept	1.43	0.08	<.001***	
	After Free	−0.18	0.06	.002**	0.29
	After Training	−0.16	0.06	.007**	0.27
	Sport Type	−0.03	0.08	.70	
PSAS (Somatic)	Intercept	1.45	0.07	<.001***	
	After Free	−0.16	0.06	.004**	0.27
	After Training	−0.14	0.06	.013*	0.27
	Sport Type	−0.08	0.07	.27	
Subj. Sleep Quality	Intercept	5.27	0.24	<.001***	
	After Free	1.10	0.15	<.001***	0.65
	After Training	1.17	0.14	<.001***	0.76
	Sport Type	0.38	0.24	.12	
Caffeine	Intercept	2.07	0.10	<.001***	
	After Free	−0.24	0.05	<.001***	0.27
	After Training	−0.18	0.04	<.001***	0.19
	Sport Type	0.15	0.12	.22	
Alcohol	Intercept	1.10	0.09	<.001***	
	After Free	0.07	0.05	.16	
	After Training	−0.07	0.05	.12	
	Sport Type	0.02	0.03	.48	

Note: *$p < .05$, **$p < .01$, ***$p < .001$. SD = Sleep diary data, Acti = Actigraphy data.

4.2. Sleep after competitions

Consistent with prior research (Caia et al., 2017; Fowler et al., 2014; Fowler et al., 2016; Fullagar et al., 2016; Sargent & Roach, 2016; Sargent et al., 2022; Shearer et al., 2015), sleep following a competition was significantly degraded compared to sleep following free and training days. Athletes went to sleep later, slept less, had lower sleep efficiency, experienced more sleep disruptions, and reported poorer subjective sleep quality after competitions. Importantly, these results were unique to nights following competitions suggesting that poor sleep is likely to be specific to the context of competition days, including factors such as kick-off times, travel times, and the high intensity, strenuous load that athletes experience during competitions. Further, potential injury and muscle pain that is more

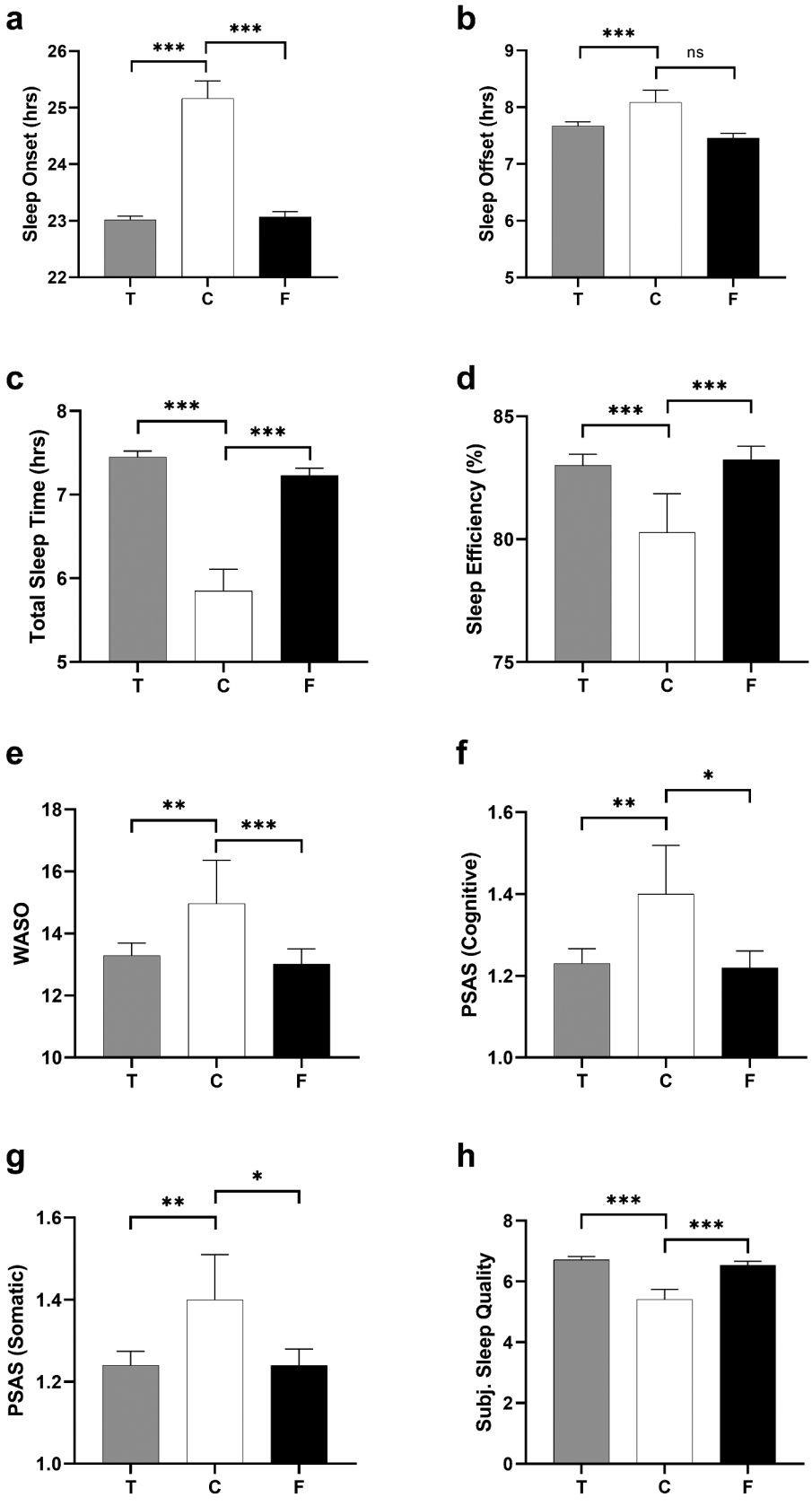

Figure 3. Bar graphs comparing the means for sleep *after* competition days, free days, and training days for (a) Sleep onset; (b) Sleep offset; (c) Total sleep time; (d) Sleep efficiency; (e) WASO = wake after sleep onset; (f) pre-sleep cognitive arousal; (g) pre-sleep somatic arousal; (h) Subjective sleep quality. Note that the comparison between raw means for sleep offset (b) after competition days and after free days appear significantly different, however after controlling for individual participant variance as random effects in the mixed model, the difference is nonsignificant. Error bars represent 95% CI. *$p < .05$, **$p < .01$, ***$p < .001$, T = after training days, C = after competition days, F = after free days.

Table 5. Linear mixed model analysis of sleep variables comparing nights after day competitions and nights after evening competitions.

Variable	Fixed Effects	Estimate	St. Error	p	d
Sleep Onset (SD)	Intercept	23.93	0.33	<.001***	
	After Night	0.72	0.28	.013*	0.43
	Sport Type	0.88	0.35	.01*	
Sleep Onset (Acti)	Intercept	24.17	0.72	.001**	
	After Night	0.54	0.39	.17	
	Sport Type	0.64	0.48	.18	
Sleep Offset (SD)	Intercept	7.18	0.24	.002**	
	After Night	0.02	0.19	.91	
	Sport Type	1.06	0.23	<.001***	
Sleep Offset (Acti)	Intercept	7.78	0.25	<.001***	
	After Night	−0.24	0.25	.33	
	Sport Type	0.97	0.28	.001**	
TST (SD)	Intercept	6.84	0.36	<.001***	
	After Night	−0.62	0.34	.07	
	Sport Type	−0.03	0.36	.93	
TST (Acti)	Intercept	6.44	0.60	<.001***	
	After Night	−0.66	0.33	.049*	0.27
	Sport Type	0.22	0.38	.56	
SE (SD)	Intercept	75.98	2.31	<.001***	
	After Night	4.69	2.39	.052	
	Sport Type	−0.67	2.16	.76	
SE (Acti)	Intercept	80.90	2.25	.06	
	After Night	−0.69	2.10	.74	
	Sport Type	0.16	2.07	.94	
WASO (SD)	Intercept	1.92	0.46	<.001***	
	After Night	−0.33	0.47	.48	
	Sport Type	0.21	0.44	.64	
WASO (Acti)	Intercept	14.56	1.72	<.001***	
	After Night	0.66	1.82	.72	
	Sport Type	−0.03	2.82	.99	
PSAS (Cognitive)	Intercept	1.55	0.16	<.001***	
	After Night	−0.16	0.15	.29	
	Sport Type	−0.17	0.18	.36	
PSAS (Somatic)	Intercept	1.65	0.15	<.001***	
	After Night	−0.22	0.15	.15	
	Sport Type	−0.30	0.16	.07	
Subj. Sleep Quality	Intercept	5.45	0.45	<.001***	
	After Night	0.18	0.31	.55	
	Sport Type	−0.17	0.53	.74	
Caffeine	Intercept	2.17	0.16	<.001***	
	After Night	−0.01	0.09	.95	
	Sport Type	−0.06	0.20	.75	
Alcohol	Intercept	1.17	0.14	.007	
	After Night	−0.04	0.12	.73	
	Sport Type	0.09	0.11	.46	

Note: *$p < .05$, **$p < .01$, ***$p < .001$. SD = Sleep diary data, Acti = Actigraphy data.

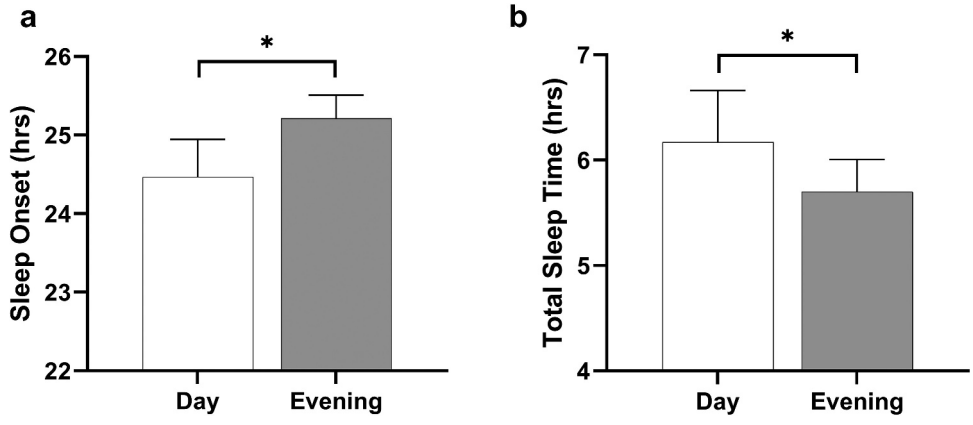

Figure 4. Bar graphs comparing the means for sleep the night after day competitions and evening competitions for (a) Sleep onset and (b) Total sleep time. Error bars represent 95% CI. *$p < .05$, **$p < .01$, ***$p < .001$.

commonly associated with competitions compared to trainings may also explain differences in sleep on these nights. Poor sleep after competitions may be due to various physiological and cognitive effects, including the elevation of adrenaline and cortisol levels (Fullagar et al., 2016; Garde et al., 2012). This rings especially true for athletes that compete closer to their habitual bedtime, where there is less time for adrenaline and cortisol levels to normalise before attempting sleep. Increased adrenaline and cortisol can heighten cognitive and somatic arousal, which can disrupt sleep (Gupta et al., 2017; Wuyts et al., 2012). Indeed, athletes in our study reported significantly higher somatic arousal the night after competitions. Poor sleep after competitions may also be due to heightened cognitive arousal. Athletes may ruminate about the "highs and lows" they just experienced. This preoccupation with the events of a competition may result in increased cognitive hyperarousal and consequently deter sleep (Jones & Sheffield, 2007).

Caffeine consumption also has physiological effects that can degrade sleep. Athletes in our study consumed significantly higher amounts of caffeine on competition days compared to free and training days. This is to be expected, as many athletes consume caffeine to enhance performance (Potgieter et al., 2018). However, increased caffeine consumption is associated with decreased sleep efficiency and duration, and increased awakenings (Clark & Landolt, 2017; Dunican et al., 2018; McLellan et al., 2016). While caffeine is useful for increasing short-term performance, athletes might consider consuming it at an earlier time to minimise disruptions to their sleep. Alcohol consumption was not significantly different the night after competition, suggesting that the effects of poor sleep after competition in our study are unlikely to be the result of alcohol consumption.

Muscle soreness, pain, and potential injury are also physiological effects that can disrupt sleep. Soreness and pain are often experienced by athletes after intense physical exertion such as competitions, and can cause more awakenings and difficulty falling asleep (Aloulou et al., 2020; Driller et al., 2022; Halson, 2014). Elite contact sports athletes are also more likely to experience injury during competitions, and these injuries may also affect the athlete's sleep (Hootman et al., 2007). Finally, disruptions to sleep, particularly delayed sleep timing, may be a consequence of post-competition commitments outside of the athletes' control, such as team meetings, post-match functions, media availability, and injury management.

We also observed differences in sleep between day and evening competitions. Athletes went to bed later and slept less after night competitions compared to day competitions. Elite night competitions are associated with both internal and external factors that may impair sleep. First, internal factors including the significant physiological and metabolic strain of elite competition can prolong the autonomic recovery window and delay parasympathetic reactivation (Seiler et al., 2007). Second, external factors such as light exposure, especially close to habitual bedtime (i.e., during night competitions), have acute alerting effects and can suppress melatonin production which can delay circadian timing of sleep-wake behaviours (Czeisler et al., 1986; O'Donnell et al., 2018). Furthermore, the time of day training and competitions occur can influence sleep and the circadian system. The athletes in the current study typically trained in the morning, which would require earlier wake up times. However, competition times were always in the afternoon or evening which would push sleep onset later. In both cases, this would lead to a build-up of sleep debt and cause more sleep irregularity, both of which could contribute to changes in sleep duration and quality.

Sleep after competition appears to be susceptible to physiological and cognitive effects. Interventions such as progressive muscle relaxation, temperature regulation and light therapy have shown promise in facilitating sleep following intense exercise (Datta et al., 2022; Knufinke et al., 2021; Lan et al., 2017). Implementing these interventions on a larger and more sustainable scale may be key to improving sleep after competitions and, consequently, maximising physiological and cognitive recovery.

The current study examined elite athletes from contact team sports. Research has found athletes from individual sports report more sleep problems because of greater individual pressure to perform (Juliff et al., 2015). Team sports often adhere to a more regular schedule, whereas individual athletes often compete on an irregular basis (Erlacher et al., 2011). Therefore, team athletes may also have developed better sleep routines compared to individual athletes. Future research should extend our findings to the domain of individual elite sports. Individual sports are also more likely to have earlier competition and training times compared to team sports, and therefore chronotype might play an important role (Lastella et al., 2015). Individual differences in circadian rhythms should also be considered in future research, given there are large variations in sleep behaviours and timing of peak performance (Facer-Childs & Brandstaetter, 2015; Facer-Childs et al., 2018). The sample size, duration of data collection, and access to an elite athlete population were significant strengths of this study. Objective and subjective sleep data were collected from 85 elite contact sports athletes in 3 timepoints over the course of a year with a total of 1,808 nights. This is significantly larger than most other studies examining the association between elite athletes' sleep during competitions (Gupta et al., 2017).

4.3. Limitations

Although our study showed significant disruptions to sleep and subjective sleep quality, we did not assess subjective sleep complaints explicitly. Future research could make use of scales such as the Athlete Sleep Behavior Questionnaire (ASBQ) (Driller et al., 2018) as well as design new athlete specific tools to shed light on the various subjective concerns of elite athletes. Another limitation of the study was the sensitivity of the actigraphy devices. Although actigraphy has been validated as a suitable alternative to PSG (Sargent et al., 2016), analysis typically underestimates sleep onset latency (SOL) and overestimates WASO. This could be a reason for our findings showing no differences in objective sleep efficiency for nights before competitive but a significant difference in subjective sleep quality. Future research could incorporate PSG or more sensitive measures of SOL. Finally, we did not measure performance or recovery as downstream consequences of disrupted sleep prior to or following competitions. Much of the sports science literature linking sleep to performance and recovery has been conducted in controlled settings (e.g., sleep restriction

and deprivation studies) and thus may lack ecological validity (Fullagar et al., 2015; Skein et al., 2011, 2013). Although the real-world setting of the current research is a step in the right direction, future studies should build upon these results by also assessing the downstream consequences of disrupted sleep on subjective and objective measures of performance and recovery in ecologically valid settings (e.g., in-season elite competition).

4.4. Implications

The results for the current study have considerable theoretical and practical implications which could benefit athletes both at elite and sub-elite levels. The understanding that sleep can be optimised before competitions is significant for athletes and sports organisations. For athletes, this is an opportunity for them to utilise specific sleep strategies to unlock the potential for competitive success. For sports organisations more broadly, this may encourage them to employ sleep coaches and commit more resources to improve athletes' sleep.

The knowledge that sleep is negatively affected after competitions could help athletes and sporting organisations prioritise strategies to mitigate these effects. For example, mindfulness is a low-cost technique that could potentially improve sleep quality among athletes (Li et al., 2018). Similarly, sports governing bodies may use the results of the current study to make informed choices on schedules, such that athlete sleep timing is less dictated by travel and competition timing.

Individual traits and differences in phenotypes may also play a role in an athlete's sleep (Horváth et al., 2016; van de Laar et al., 2010). For example, athletes high in trait anxiety may be more prone to nervousness and worrying, and this in turn may result in difficulty falling asleep due to cognitive and somatic arousal. It is possible that traits contributing towards success and selection within elite sports (e.g., athletes high in perfectionism) may indirectly predispose athletes to poorer sleep and pre-sleep conditions. Future research could consider looking at the potential moderating effects of trait anxiety on pre- and post-competition arousal, and how it may consequently affect sleep.

5. Conclusion

This study examined elite contact sports athlete sleep before and after competitions. We found that sleep improved during the nights before a competition, suggesting that athletes were prepared and took steps to mitigate the effects of pre-competition arousal. However, sleep was negatively affected the nights after a competition and especially the night after an evening competition, which may be due to the physiological, psychological, and circadian effects of a competing at the highest level. These results provide valuable knowledge for elite athletes, and shed light on the importance of prioritising sleep before and, more importantly, after competition. These findings also suggest a need for expert sleep consultants to provide tailored recommendations to optimise sleep.

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The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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ORCID

Joshua Leota http://orcid.org/0000-0002-7714-4630
Elise R. Facer-Childs http://orcid.org/0000-0003-4549-9266

Author contributions

JES, JL, and ERF-C contributed to the conception and design of the study and to manuscript revisions. All authors contributed to the data collection. JES and JL analysed the data. JES and JL wrote the manuscript, with contributions from ERF-C. All authors read and approved the submitted version.

Data availability statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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